

Initiating Search

November 10, 2025, 2:16 PM

• Search:

CAS Lexicon Search:

Swern oxidation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (594)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (784)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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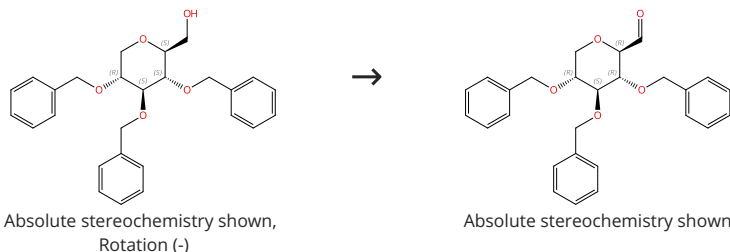
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-10396252

Steps: 1 Yield: 100%

Synthesis of aromatic C-xylosides by position inversion of glucose

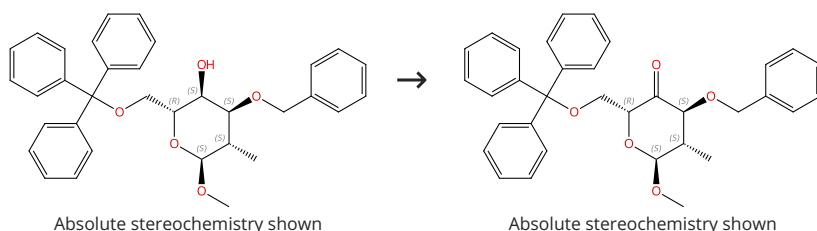
By: Malmberg, Jesper; et al

Bioorganic & Medicinal Chemistry (2006), 14(19), 6659-6665.

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; rt → -61 °C; 5 min, -61 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -61 °C
- 1.3 **Reagents:** Triethylamine; 15 min, -61 °C; -61 °C → rt
- 1.4 **Solvents:** Water; rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-4075565

Steps: 1 Yield: 100%

An approach to the total synthesis of the Prelog-Djerassi lactone

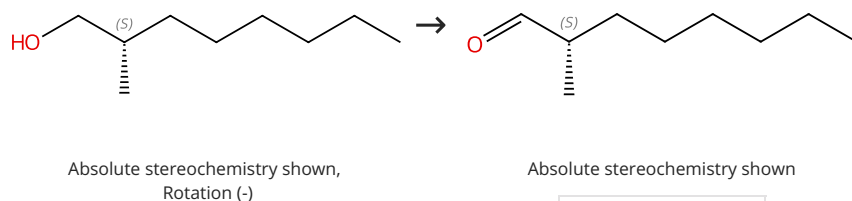
By: Jones, Keith; et al

Journal of the Chemical Society, Perkin Transactions 1: Organic and Bio-Organic Chemistry (1972-1999) (1987), (3), 537-45.

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

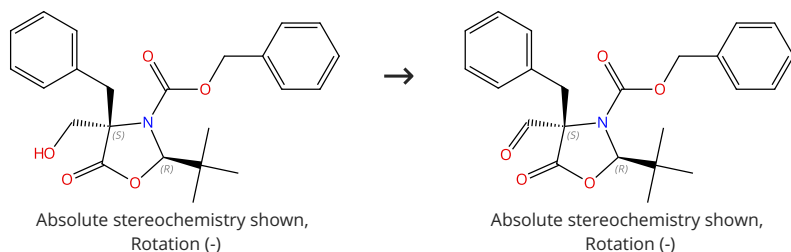

 Suppliers (8)

 Suppliers (2)

31-480-CAS-15309429	Steps: 1 Yield: 100%	Total synthesis of emericellamides A and B
1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride Solvents: Dichloromethane; -78 °C; 1 h, -78 °C	Experimental Protocols	By: Li, Shuo; et al Tetrahedron (2009), 65(13), 2695-2702.
1.2 Solvents: Dichloromethane; -78 °C; 30 min, -78 °C		
1.3 Reagents: Diisopropylethylamine; 30 min, -78 °C; 20 min, -50 °C		

Scheme 4 (1 Reaction)

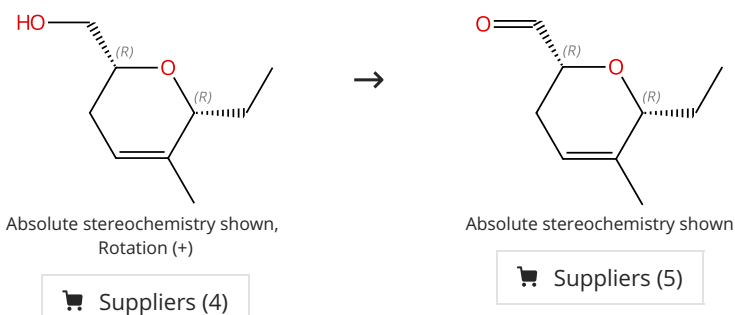
Steps: 1 Yield: 100%



31-480-CAS-12052665	Steps: 1 Yield: 100%	Phe-Ala-based diazaspirocyclic lactam as nucleator of type II' β-Turn
1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride Solvents: Dichloromethane; -78 °C; 10 min, -78 °C	Experimental Protocols	By: Sacchetti, Alessandro; et al Journal of Organic Chemistry (2011), 76(3), 833-839.
1.2 Solvents: Dichloromethane; 1.5 h, -78 °C		
1.3 Reagents: Triethylamine; -78 °C; -78 °C $\rightarrow$ rt; 2 h, rt		

Scheme 5 (1 Reaction)

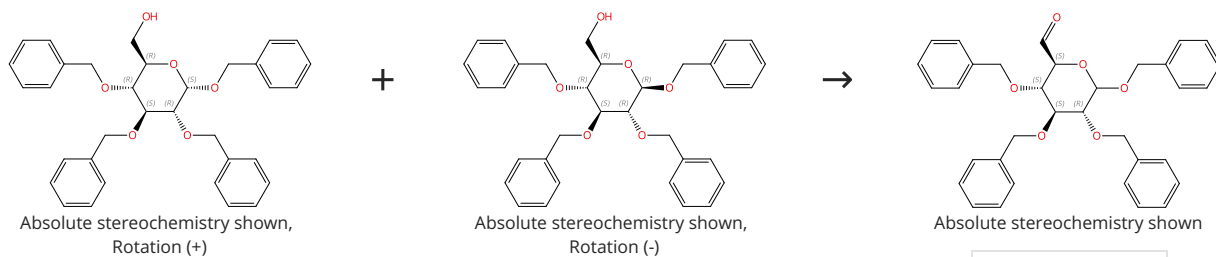
Steps: 1 Yield: 100%



31-480-CAS-4152930	Steps: 1 Yield: 100%	Total Synthesis of (+)-Ambruticin S: Probing the Pharmacophoric Subunit
1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride Solvents: Dichloromethane; 15 min, -78 °C	Experimental Protocols	By: Hanessian, Stephen; et al Journal of Organic Chemistry (2010), 75(16), 5601-5618.
1.2 Solvents: Dichloromethane; 1 h, -78 °C		
1.3 Reagents: Triethylamine; 15 min, -78 °C; -78 °C $\rightarrow$ 0 °C; 15 min, 0 °C		

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (49)

Suppliers (53)

Supplier (1)

31-480-CAS-8199070

Steps: 1 Yield: 100%

Synthesis of cyclophospho-glucoses and glucitols

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 40 min, -78 °C
- 1.2 **Solvents:** Dichloromethane; 2 h, -78 °C
- 1.3 **Reagents:** Triethylamine; 10 min, -78 °C; -78 °C → rt
- 1.4 **Solvents:** Water; rt

Experimental Protocols

By: Amigues, E. J.; et al

Tetrahedron (2007), 63(40), 10042-10053.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-13634133

Steps: 1 Yield: 100%

Intramolecular Diels-Alder reactions using chiral ruthenium Lewis acids and application in the total synthesis of ent-ledol

- 1.1 **Reagents:** Manganese oxide (MnO₂)
Solvents: Diethyl ether; 30 min, rt

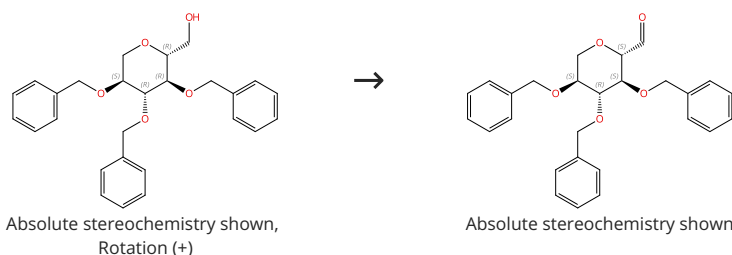
Experimental Protocols

By: Thamapipol, Sirinporn; et al

Organic & Biomolecular Chemistry (2011), 9(21), 7564-7570.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

31-480-CAS-12512697

Steps: 1 Yield: 100%

Synthesis of aromatic C-xylosides by position inversion of glucose

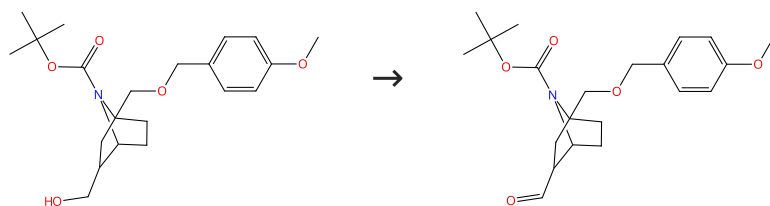
- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; rt → -61 °C; 5 min, -61 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -61 °C
- 1.3 **Reagents:** Triethylamine; 15 min, -61 °C; -61 °C → rt
- 1.4 **Solvents:** Water; rt

By: Malmberg, Jesper; et al

Bioorganic & Medicinal Chemistry (2006), 14(19), 6659-6665.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



31-480-CAS-10773520

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 20 min, -78 °C
- 1.2 **Solvents:** Dichloromethane; 40 min, -78 °C
- 1.3 **Reagents:** Triethylamine; -78 °C → rt; 1 h, rt
- 1.4 **Reagents:** Water; rt

Experimental Protocols

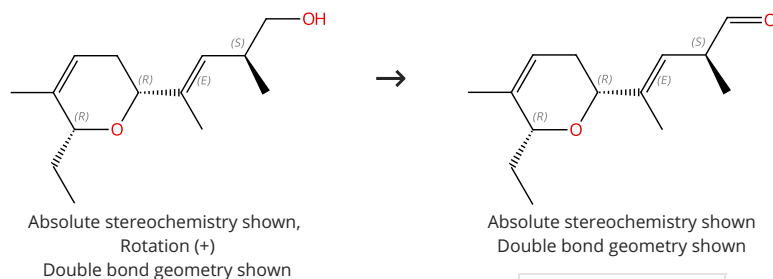
Water-Stable Helical Structure of Tertiary Amides of Bicyclic β -Amino Acid Bearing 7-Azabicyclo[2.2.1]heptane. Full Control of Amide Cis-Trans Equilibrium by Bridgehead Substitution

By: Hosoya, Masahiro; et al

Journal of the American Chemical Society (2010), 132(42), 14780-14789.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

31-480-CAS-12664815

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium bicarbonate, Dess-Martin periodinane
Solvents: Dichloromethane; 3 h, 0 °C
- 1.2 **Reagents:** Sodium bicarbonate, Sodium thiosulfate
Solvents: Water; 0 °C

Experimental Protocols

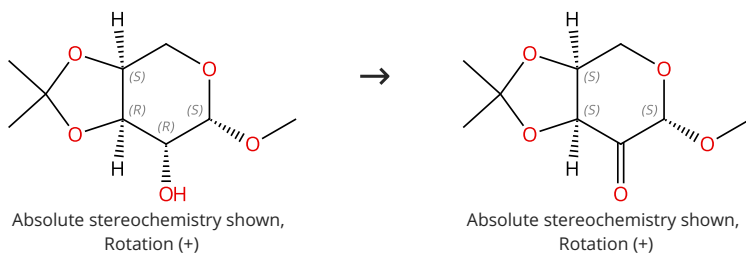
Total Synthesis of (+)-Ambruticin S: Probing the Pharmacophoric Subunit

By: Hanessian, Stephen; et al

Journal of Organic Chemistry (2010), 75(16), 5601-5618.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (15)

Suppliers (17)

31-480-CAS-6613698

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane
- 1.2 **Reagents:** Triethylamine

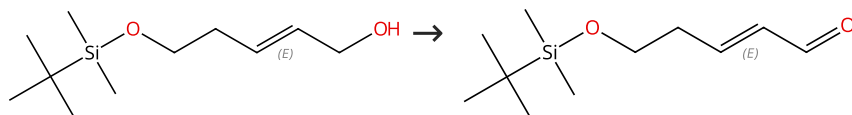
A practical synthesis of L-ribose

By: Akagi, Masao; et al

Chemical & Pharmaceutical Bulletin (2002), 50(6), 866-868.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Suppliers (7)

Suppliers (4)

31-480-CAS-10863836

Steps: 1 Yield: 100%

1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride, Triethylamine
Solvents: Dichloromethane; 1.5 h, -78 °C

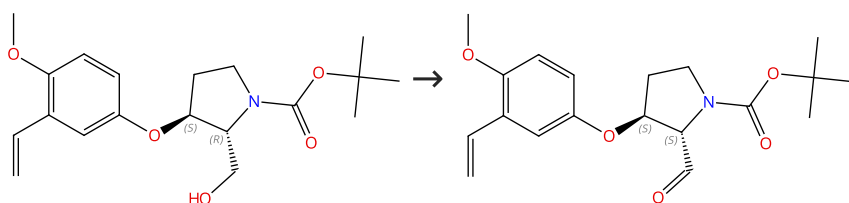
Studies on the total synthesis of amphidinolide O (II): A stereoselective synthesis of C1-C11 fragment

By: Jang, Mi-Yeon; et al

Bulletin of the Korean Chemical Society (2005), 26(10), 1497-1498.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

31-480-CAS-10439913

Steps: 1 Yield: 100%

1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 30 min, -78 °C
 1.2 **Solvents:** Dichloromethane; -78 °C; 30 min, -78 °C
 1.3 **Reagents:** Triethylamine; -78 °C; 4 h, -78 °C → 0 °C
 1.4 **Reagents:** Water; 0 °C

Experimental Protocols

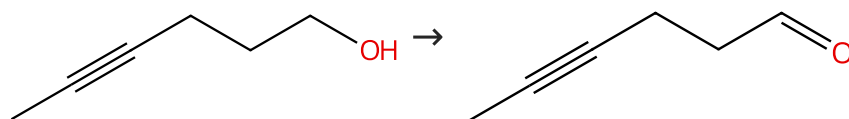
Total synthesis of the cyclopeptide alkaloid Paliurine E. Insights into macrocyclization by ene-enamide RCM

By: Toumi, Mathieu; et al

Journal of Organic Chemistry (2008), 73(4), 1270-1281.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (37)

Suppliers (15)

31-480-CAS-2221353

Steps: 1 Yield: 100%

1.1 **Reagents:** Oxalyl chloride
Solvents: Dimethyl sulfoxide, Dichloromethane; -50 °C; 20 min, -50 °C
 1.2 **Solvents:** Dichloromethane; 30 min, -50 °C
 1.3 **Reagents:** Triethylamine; -50 °C → rt; 2 h, rt

Experimental Protocols

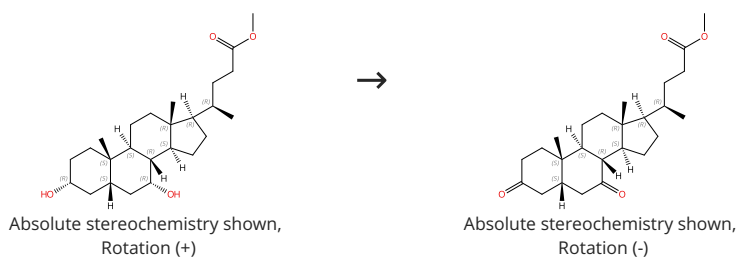
Identification and Synthesis of Macrolide Pheromones of the Grain Beetle *Oryzaephilus surinamensis* and the Frog *Spinomantis aglavei*

By: Hoetling, Susann; et al

Chemistry - A European Journal (2014), 20(11), 3183-3191.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

Suppliers (11)

31-614-CAS-34334478

Steps: 1 Yield: 100%

Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups

- 1.1 **Reagents:** Chromium trioxide
Solvents: Acetone, Water; 0 °C; 30 min, 0 °C
- 1.2 **Reagents:** Isopropanol

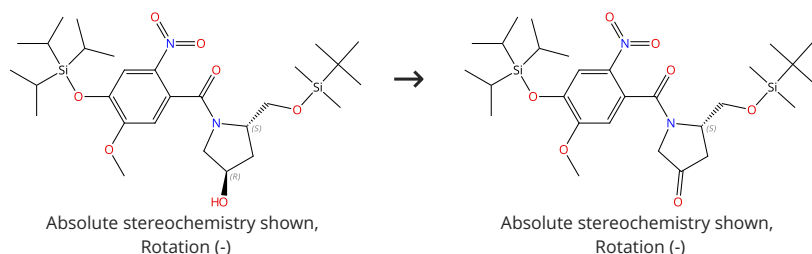
By: Kaspar, Miroslav; et al

Journal of Organic Chemistry (2022), 87(14), 9157-9170.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (30)

31-480-CAS-16353186

Steps: 1 Yield: 100%

Design and Synthesis of Tesirine, a Clinical Antibody- Drug Conjugate Pyrrolobenzodiazepine Dimer Payload

- 1.1 **Reagents:** Trichloroisocyanuric acid
Catalysts: Tempo
Solvents: Dichloromethane; 20 min, 0 °C

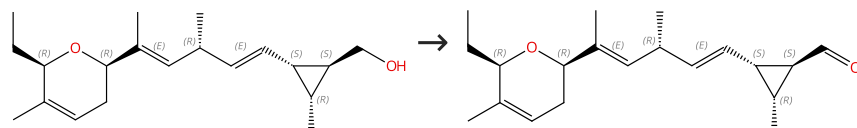
By: Tiberghien, Arnaud C.; et al

ACS Medicinal Chemistry Letters (2016), 7(11), 983-987.

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shown

Absolute stereochemistry shown
Double bond geometry shown

Suppliers (2)

31-480-CAS-15366506

Steps: 1 Yield: 100%

Total Synthesis of (+)-Ambruticin S: Probing the Pharmacophoric Subunit

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 20 min, -78 °C
- 1.2 **Solvents:** Dichloromethane; 1 h, -78 °C
- 1.3 **Reagents:** Triethylamine; 15 min, -78 °C; -78 °C → rt

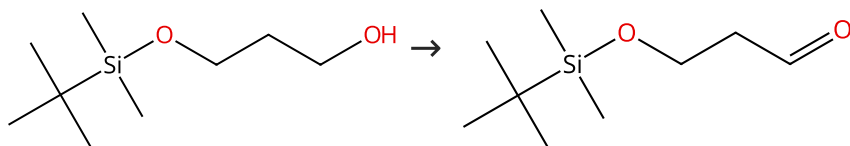
By: Hanessian, Stephen; et al

Journal of Organic Chemistry (2010), 75(16), 5601-5618.

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (63)

Suppliers (68)

31-480-CAS-8895052

Steps: 1 Yield: 100%

Studies on the total synthesis of amphidinolide O (II): A stereoselective synthesis of C1-C11 fragment

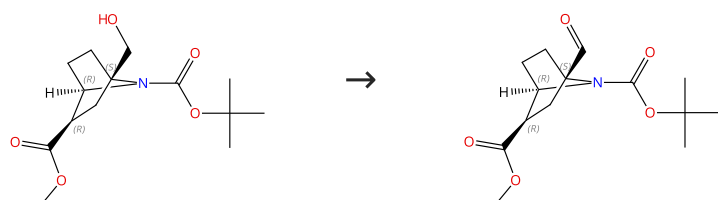
By: Jang, Mi-Yeon; et al

Bulletin of the Korean Chemical Society (2005), 26(10), 1497-1498.

1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride, Triethylamine
Solvents: Dichloromethane; 1.5 h, -78 °C

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Relative stereochemistry shown

Relative stereochemistry shown

31-480-CAS-1611114

Steps: 1 Yield: 100%

Water-Stable Helical Structure of Tertiary Amides of Bicyclic β -Amino Acid Bearing 7-Azabicyclo[2.2.1]heptane. Full Control of Amide Cis-Trans Equilibrium by Bridgehead Substitution

By: Hosoya, Masahiro; et al

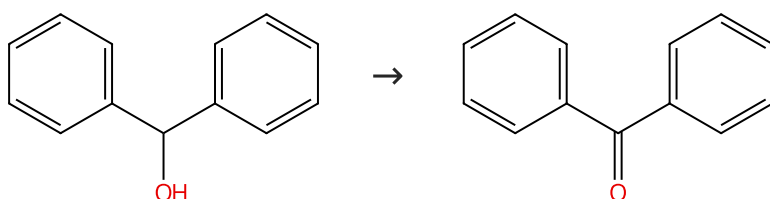
Journal of the American Chemical Society (2010), 132(42), 14780-14789.

1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 20 min, -78 °C
 1.2 **Solvents:** Dichloromethane; 40 min, -78 °C
 1.3 **Reagents:** Triethylamine; -78 °C; -78 °C $\rightarrow$ rt; 1 h, rt
 1.4 **Reagents:** Water; rt

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (97)

Suppliers (136)

31-480-CAS-6057779

Steps: 1 Yield: 100%

New odorless method for the Corey-Kim and Swern oxidations utilizing dodecyl methyl sulfide (Dod-S-Me)

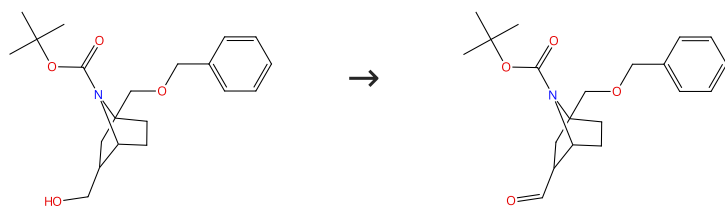
By: Ohsugi, Shin-ichi; et al

Tetrahedron (2003), 59(42), 8393-8398.

1.1 **Reagents:** Chlorosuccinimide, 1-(Methylthio)dodecane
Solvents: Acetone; 30 min, -40 °C
 1.2 **Solvents:** Dichloromethane; 2 h, -40 °C
 1.3 **Reagents:** Triethylamine; 2.5 h, -40 °C
 1.4 **Solvents:** Water
 1.5 **Reagents:** Hydrochloric acid
Solvents: Water; neutralized

Scheme 21 (1 Reaction)

Steps: 1 Yield: 99%



31-480-CAS-1250574

Steps: 1 Yield: 99%

Water-Stable Helical Structure of Tertiary Amides of Bicyclic β -Amino Acid Bearing 7-Azabicyclo[2.2.1]heptane. Full Control of Amide Cis-Trans Equilibrium by Bridgehead Substitution

By: Hosoya, Masahiro; et al

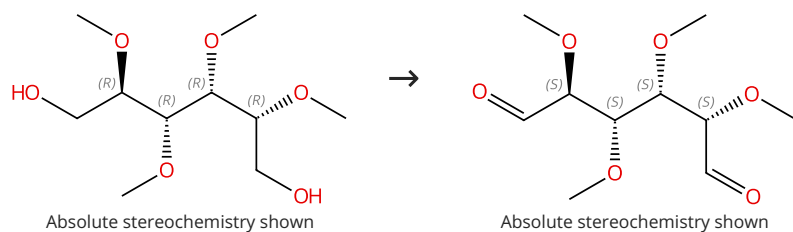
Journal of the American Chemical Society (2010), 132(42), 14780-14789.

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 20 min, -78 °C
- 1.2 **Solvents:** Dichloromethane; 40 min, -78 °C
- 1.3 **Reagents:** Triethylamine; -78 °C; -78 °C → rt; 1 h, rt
- 1.4 **Reagents:** Water; rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 99%



31-480-CAS-6994427

Steps: 1 Yield: 99%

Synthesis of allo- and epi-Inositol via the NHC-Catalyzed Carbocyclization of Carbohydrate-Derived Dialdehydes

By: Stockton, Kieran P.; et al

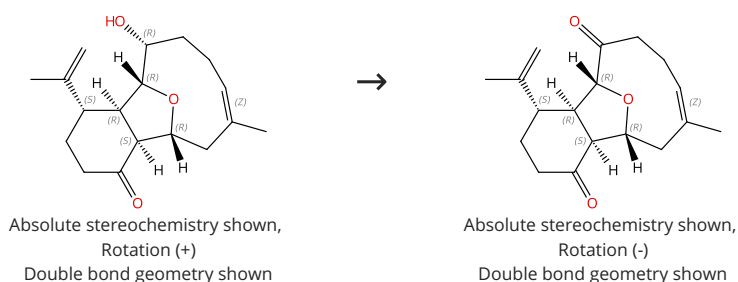
Journal of Organic Chemistry (2014), 79(11), 5088-5096.

- 1.1 **Reagents:** Oxalyl chloride
Solvents: Dimethyl sulfoxide, Dichloromethane; -78 °C; 10 min, < -60 °C; 20 min, -60 °C
- 1.2 **Reagents:** Triethylamine; 15 min, < -60 °C; 1.5 h, -60 °C → rt

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 99%



31-480-CAS-4727630

Steps: 1 Yield: 99%

Total Synthesis of the Proposed Structure of Briarellin J

By: Crimmins, Michael T.; et al

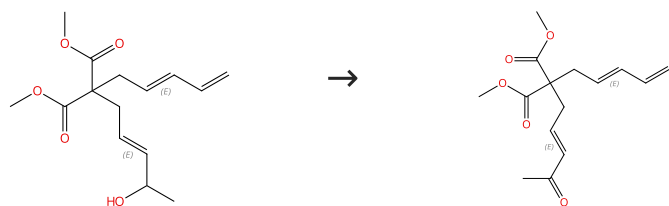
Organic Letters (2010), 12(21), 5028-5031.

- 1.1 **Reagents:** Pyridine, Dess-Martin periodinane
Solvents: Dichloromethane; 30 min, rt
- 1.2 **Reagents:** Sodium bicarbonate, Sodium thiosulfate
Solvents: Water; rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Double bond geometry shown

31-480-CAS-9369558

Steps: 1 Yield: 99%

Intramolecular Diels-Alder reactions using chiral ruthenium Lewis acids and application in the total synthesis of ent-ledol

By: Thamapipol, Sirinporn; et al

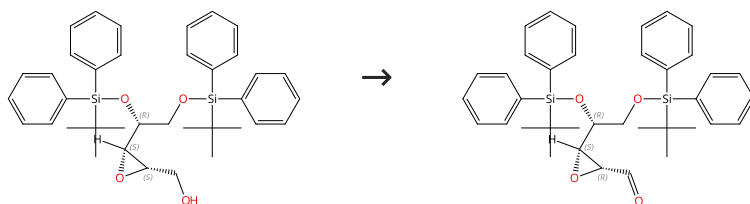
Organic & Biomolecular Chemistry (2011), 9(21), 7564-7570.

1.1 Reagents: Manganese oxide (MnO₂)
Solvents: Diethyl ether; 30 min, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

31-480-CAS-964478

Steps: 1 Yield: 99%

Direct access to furanosidic eight-membered ulosonic esters from cis- α,β -epoxy aldehydes

By: Sugisaki, Claudia H.; et al

European Journal of Organic Chemistry (2003), (4), 672-688.

1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 10 min, -78 °C

1.2 Solvents: Dichloromethane; 1 h, -78 °C

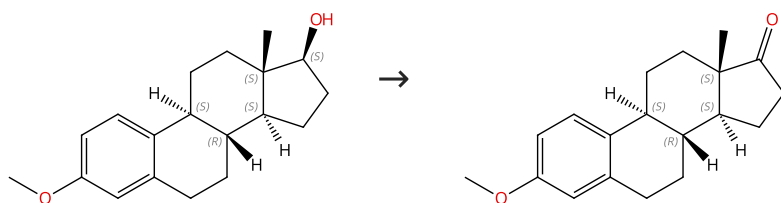
1.3 Reagents: Triethylamine; -78 °C → rt; 15 min, rt

1.4 Solvents: Water; 10 min, rt

1.5 Reagents: Sodium bicarbonate
Solvents: Water; rt

Scheme 26 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

Absolute stereochemistry shown,
Rotation (+)

Suppliers (39)

Suppliers (68)

31-480-CAS-9082659

Steps: 1 Yield: 99%

A practical improvement of odorless Corey-Kim and Swern oxidations

By: Nishide, Kiyoharu; et al

Green Chemistry (2004), 6(3), 142-146.

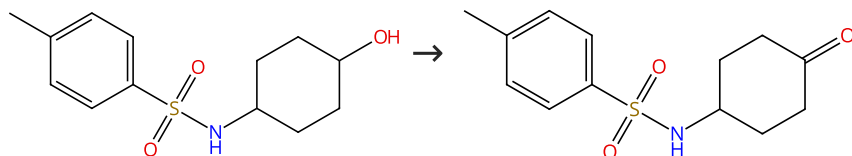
1.1 Reagents: Chlorosuccinimide, 4-[6-(Methylthio)hexyl] morpholine
Solvents: Acetonitrile; 30 min, -40 °C

1.2 Solvents: Acetonitrile; 2 h, -40 °C

1.3 Reagents: Triethylamine; 2.5 h, -40 °C; 8 h, -40 °C → rt

Scheme 27 (2 Reactions)

Steps: 1 Yield: 99%



Suppliers (5)

Suppliers (3)

31-480-CAS-13300015

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-Imidazolium, 1-methyl-3-[10-(methylsulfinyl)decyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols

Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides

By: Tsuchiya, Daisuke; et al

Tetrahedron (2012), 68(34), 6849-6855.

31-480-CAS-11167913

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-Imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols

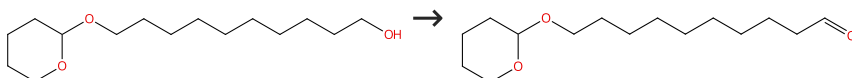
Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides

By: Tsuchiya, Daisuke; et al

Tetrahedron (2012), 68(34), 6849-6855.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (4)

Suppliers (3)

31-480-CAS-9106479

Steps: 1 Yield: 99%

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride, Triethylamine
Solvents: Dichloromethane; -78 °C

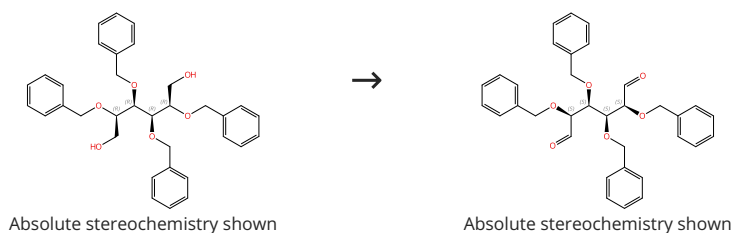
Synthesis of fluorescence-labeled sphingosine and sphingosine 1-phosphate; effective tools for sphingosine and sphingosine 1-phosphate behavior

By: Hakogi, Toshikazu; et al

Bioorganic & Medicinal Chemistry Letters (2003), 13(4), 661-664.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (3)

31-480-CAS-4846055

Steps: 1 Yield: 99%

Synthesis of allo- and epi-Inositol via the NHC-Catalyzed Carbocyclization of Carbohydrate-Derived Dialdehydes

By: Stockton, Kieran P.; et al

Journal of Organic Chemistry (2014), 79(11), 5088-5096.

1.1 Reagents: Oxalyl chloride

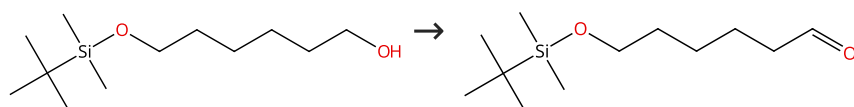
Solvents: Dimethyl sulfoxide, Dichloromethane; -78 °C; 10 min, < -60 °C; 20 min, -60 °C

1.2 Reagents: Triethylamine; 15 min, < -60 °C; 1.5 h, -60 °C → rt

Experimental Protocols

Scheme 30 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (40)

Suppliers (31)

31-480-CAS-629346

Steps: 1 Yield: 99%

Investigation of the origin and synthetic application of the pseudodilution effect for Pd-catalyzed macrocyclisations in concentrated solutions with immobilized catalysts

By: Brehm, Elisabeth; et al

Organic & Biomolecular Chemistry (2013), 11(29), 4750-4756.

1.1 Reagents: Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; -78 °C; 10 min, -78 °C

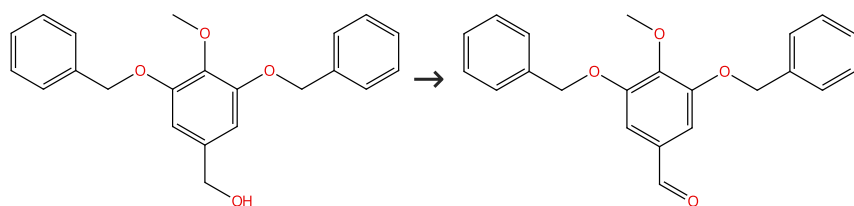
1.2 Solvents: Dichloromethane; -78 °C; 1.5 h, -78 °C

1.3 Reagents: Triethylamine; overnight, -78 °C → rt

Experimental Protocols

Scheme 31 (3 Reactions)

Steps: 1 Yield: 99%



Suppliers (3)

Suppliers (4)

31-480-CAS-4042758

Steps: 1 Yield: 99%

Development of Odorless Thiols and Sulfides and Their Applications to Organic Synthesis

By: Nishide, Kiyoharu; et al

Monatshefte fuer Chemie (2004), 135(2), 189-200.

1.1 Reagents: Chlorosuccinimide, 1-(Methylthio)dodecane
Solvents: Dichloromethane; -40 °C; 30 min, -40 °C

1.2 Solvents: Dichloromethane; -40 °C; 2 h, -40 °C

1.3 Reagents: Triethylamine; -40 °C; 14 h, -40 °C

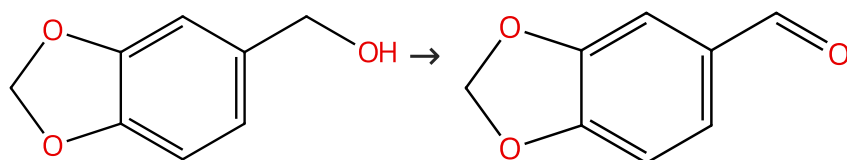
1.4 Reagents: Hydrochloric acid
Solvents: Water; neutralized

Experimental Protocols

31-480-CAS-2070092 Steps: 1 Yield: 99% 1.1 Reagents: Chlorosuccinimide, 1-(Methylthio)dodecane Solvents: Dichloromethane; 30 min, -40 °C 1.2 Solvents: Dichloromethane; 2 h, -40 °C 1.3 Reagents: Triethylamine; 14 h, -40 °C 1.4 Solvents: Water 1.5 Reagents: Hydrochloric acid Solvents: Water; neutralized	New odorless method for the Corey-Kim and Swern oxidations utilizing dodecyl methyl sulfide (Dod-S-Me) By: Ohsugi, Shin-ichi; et al Tetrahedron (2003), 59(42), 8393-8398.
31-480-CAS-13340379 Steps: 1 Yield: 99% 1.1 Reagents: Triethylamine, Chlorosuccinimide, 1-(Methylthio)dodecane Solvents: Dichloromethane; 30 min, -40 °C; 2 h, -40 °C; 14 h, -40 °C 1.2 Reagents: Hydrochloric acid Solvents: Water	New odorless protocols for the Swern and Corey-Kim oxidations By: Nishide, Kiyoharu; et al Tetrahedron Letters (2002), 43(29), 5177-5179.

Scheme 32 (2 Reactions)

Steps: 1 Yield: 99%



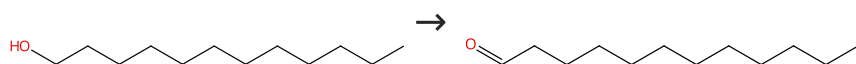
Suppliers (80)

Suppliers (42)

31-480-CAS-9084780 Steps: 1 Yield: 99% 1.1 Reagents: Oxalyl chloride, 1<i>H</i>-Imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; 30 min, -70 °C 1.2 Solvents: Dichloromethane; 30 min, -70 °C 1.3 Reagents: Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 6 - 7, rt	Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides By: Tsuchiya, Daisuke; et al Tetrahedron (2012), 68(34), 6849-6855.
Experimental Protocols	
31-480-CAS-2658239 Steps: 1 Yield: 99% 1.1 Reagents: Oxalyl chloride, 1<i>H</i>-Imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.2 Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.3 Reagents: Triethylamine; -70 °C; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 6 - 7, rt	Swern oxidation of alcohols with ion-supported methyl sulfoxide and oxalyl chloride By: Tsuchiya, Daisuke; et al Synlett (2011), (18), 2701-2704.
Experimental Protocols	

Scheme 33 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (99)

Suppliers (76)

31-480-CAS-10044900

Steps: 1 Yield: 99%

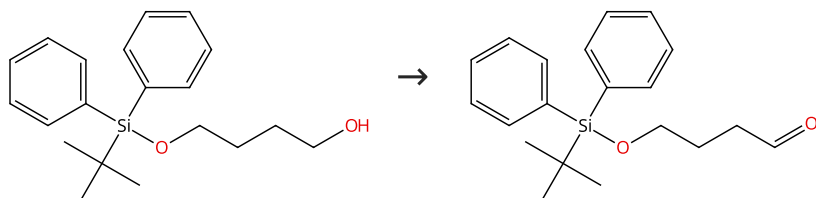
Modification of the Swern oxidation: use of a soluble polymer-bound, recyclable, and odorless sulfoxide

- 1.1 **Reagents:** Oxalyl chloride, Triethylamine, 6-(Methylsulfinyl) hexanoic acid (chloromethylated polystyrene and polyethylene glycol supported)
Solvents: Dichloromethane

By: Harris, Joanna M.; et al
 Journal of Organic Chemistry (1998), 63(7), 2407-2409.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (22)

Suppliers (14)

31-480-CAS-9425914

Steps: 1 Yield: 99%

Stereoselective synthesis of the C(53)-C(67) polyene fragment of amphidinol-3

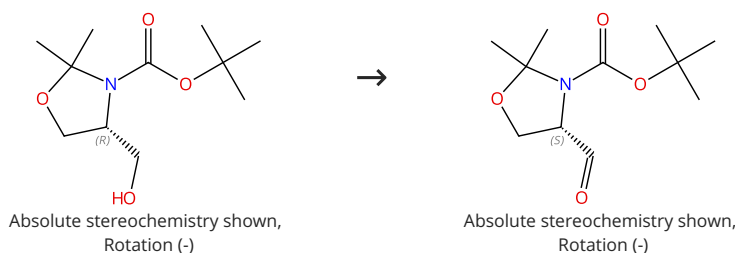
- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane; 1 h, -78 °C
 1.2 **Reagents:** Triethylamine; 30 min, -78 °C → rt

By: Colobert, Francoise; et al
 Synlett (2007), (15), 2351-2354.

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (52)

Suppliers (80)

31-480-CAS-5674612

Steps: 1 Yield: 99%

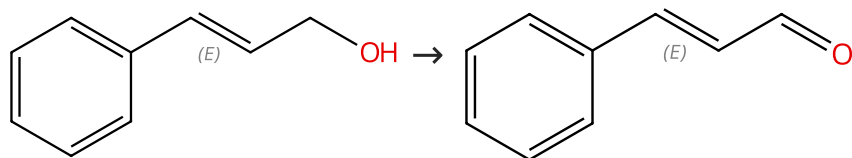
Synthesis of 1,1-dimethylethyl (S)-4-formyl-2,2-dimethyl-3-oxazolidinonecarboxylate by oxidation of the alcohol

- 1.1 **Reagents:** Dimethyl sulfoxide, Oxalyl chloride
Solvents: Dichloromethane
 1.2 **Reagents:** Diisopropylethylamine

By: Dondoni, Alessandro; et al
 Organic Syntheses (2000), 77, 64-77.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (66)

Double bond geometry shown

Suppliers (109)

31-480-CAS-6919998

Steps: 1 Yield: 99%

Swern oxidation of alcohols with ion-supported methyl sulfoxide and oxalyl chloride

By: Tsuchiya, Daisuke; et al

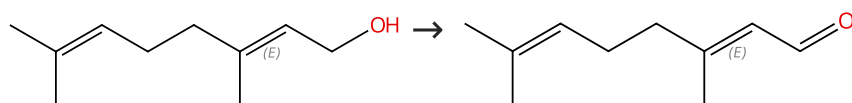
Synlett (2011), (18), 2701-2704.

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-Imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; -70 °C; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; -70 °C; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; -70 °C; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols

Scheme 37 (1 Reaction)

Steps: 1 Yield: 99%



Double bond geometry shown

Suppliers (117)

Double bond geometry shown

Suppliers (36)

31-480-CAS-4776952

Steps: 1 Yield: 99%

Swern oxidation of alcohols with ion-supported methyl sulfoxide and oxalyl chloride

By: Tsuchiya, Daisuke; et al

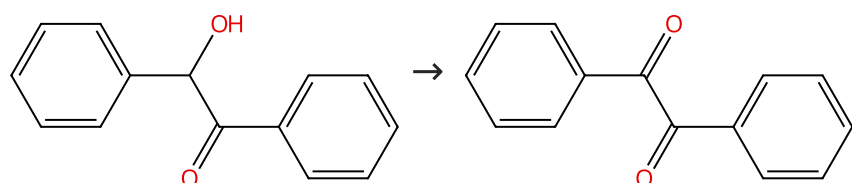
Synlett (2011), (18), 2701-2704.

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-Imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; -70 °C; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; -70 °C; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; -70 °C; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols

Scheme 38 (2 Reactions)

Steps: 1 Yield: 99%



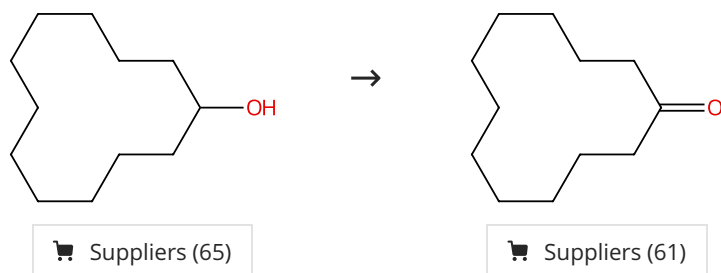
Suppliers (118)

Suppliers (101)

31-480-CAS-32658 Steps: 1 Yield: 99% 1.1 Reagents: Chlorosuccinimide, 1-(Methylthio)dodecane Solvents: Dichloromethane; -40 °C; 30 min, -40 °C 1.2 Solvents: Dichloromethane; -40 °C; 2 h, -40 °C 1.3 Reagents: Triethylamine; -40 °C; 14 h, -40 °C 1.4 Reagents: Hydrochloric acid Solvents: Water; neutralized Experimental Protocols	Development of Odorless Thiols and Sulfides and Their Applications to Organic Synthesis By: Nishide, Kiyoharu; et al Monatshefte fuer Chemie (2004), 135(2), 189-200.
31-480-CAS-32425 Steps: 1 Yield: 99% 1.1 Reagents: Chlorosuccinimide, 1-(Methylthio)dodecane Solvents: Dichloromethane; 30 min, -40 °C 1.2 Solvents: Dichloromethane; 2 h, -40 °C 1.3 Reagents: Triethylamine; 14 h, -40 °C 1.4 Solvents: Water 1.5 Reagents: Hydrochloric acid Solvents: Water; neutralized	New odorless method for the Corey-Kim and Swern oxidations utilizing dodecyl methyl sulfide (Dod-S-Me) By: Ohsugi, Shin-ichi; et al Tetrahedron (2003), 59(42), 8393-8398.

Scheme 39 (3 Reactions)

Steps: 1 Yield: 99%

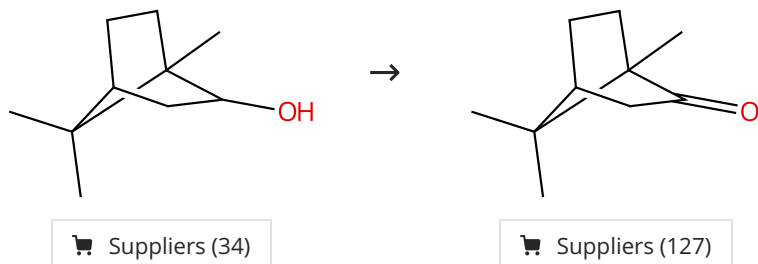


31-480-CAS-15421723 Steps: 1 Yield: 99% 1.1 Reagents: Oxalyl chloride, 1 <i>H</i> -imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; 30 min, -70 °C 1.2 Solvents: Dichloromethane; 30 min, -70 °C 1.3 Reagents: Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 6 - 7, rt Experimental Protocols	Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides By: Tsuchiya, Daisuke; et al Tetrahedron (2012), 68(34), 6849-6855.
31-480-CAS-4832221 Steps: 1 Yield: 99% 1.1 Reagents: Chlorosuccinimide, 1 <i>H</i> -imidazolium, 1-methyl-3-[6-(methylthio)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; 0.5 h, -20 °C 1.2 Reagents: Triethylamine; 14 h, -20 °C 1.3 Solvents: Water Experimental Protocols	Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides By: Tsuchiya, Daisuke; et al Tetrahedron (2012), 68(34), 6849-6855.

31-480-CAS-11439750 Steps: 1 Yield: 99% 1.1 Reagents: Oxalyl chloride, 1<i>H</i>-imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.2 Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.3 Reagents: Triethylamine; -70 °C; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 6 - 7, rt Experimental Protocols	Swern oxidation of alcohols with ion-supported methyl sulfoxide and oxalyl chloride By: Tsuchiya, Daisuke; et al Synlett (2011), (18), 2701-2704.
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Scheme 40 (1 Reaction)

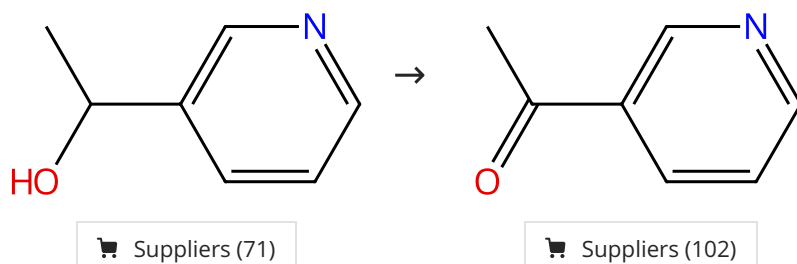
Steps: 1 Yield: 99%



31-480-CAS-166925 Steps: 1 Yield: 99% 1.1 Reagents: Oxalyl chloride, 1<i>H</i>-imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1) Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.2 Solvents: Dichloromethane; -70 °C; 30 min, -70 °C 1.3 Reagents: Triethylamine; -70 °C; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt 1.4 Reagents: Hydrochloric acid Solvents: Water; pH 6 - 7, rt Experimental Protocols	Swern oxidation of alcohols with ion-supported methyl sulfoxide and oxalyl chloride By: Tsuchiya, Daisuke; et al Synlett (2011), (18), 2701-2704.
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Scheme 41 (1 Reaction)

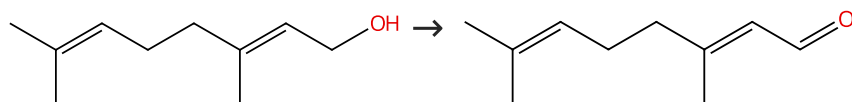
Steps: 1 Yield: 99%



31-480-CAS-12394375 Steps: 1 Yield: 99% 1.1 Reagents: Chlorosuccinimide, 1-(Methylthio)dodecane Solvents: Tetrahydrofuran; 30 min, -40 °C 1.2 Solvents: Tetrahydrofuran; 2 h, -40 °C 1.3 Reagents: Triethylamine; 14 h, -40 °C	New odorless method for the Corey-Kim and Swern oxidations utilizing dodecyl methyl sulfide (Dod-S-Me) By: Ohsugi, Shin-ichi; et al Tetrahedron (2003), 59(42), 8393-8398.
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Scheme 42 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (13)

Suppliers (87)

31-480-CAS-2079284

Steps: 1 Yield: 99%

Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides

By: Tsuchiya, Daisuke; et al

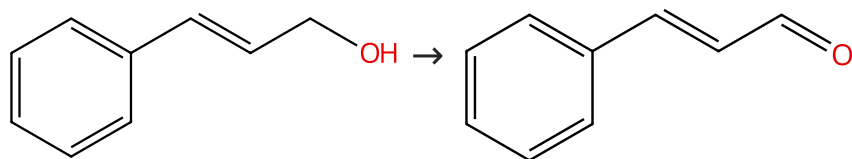
Tetrahedron (2012), 68(34), 6849-6855.

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (86)

Suppliers (52)

31-480-CAS-13618094

Steps: 1 Yield: 99%

Efficient Swern oxidation and Corey-Kim oxidation with ion-supported methyl sulfoxides and methyl sulfides

By: Tsuchiya, Daisuke; et al

Tetrahedron (2012), 68(34), 6849-6855.

- 1.1 **Reagents:** Oxalyl chloride, 1*H*-imidazolium, 1-methyl-3-[6-(methylsulfinyl)hexyl]-, 4-methylbenzenesulfonate (1:1)
Solvents: Dichloromethane; 30 min, -70 °C
- 1.2 **Solvents:** Dichloromethane; 30 min, -70 °C
- 1.3 **Reagents:** Triethylamine; 1 h, -70 °C; -70 °C → -60 °C; 1.5 h, -60 °C; -60 °C → -50 °C; 1 h, -50 °C; -50 °C → rt; 2 h, rt
- 1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 6 - 7, rt

Experimental Protocols